DAA LAB — WEEK 3
Programs 11–15
Source: extracted from your uploaded DAA lab PDF.

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PROGRAM 11: Count zeros in a sorted binary array (divide and conquer)
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```
#include <stdio.h>
int findFirstZero(int arr[], int low, int high) {
   if (low > high)
       return -1;
   int mid = low + (high - low) / 2;
   if (arr[mid] == 0) {
       if (mid == 0 || arr[mid - 1] == 1)
           return mid;
       return findFirstZero(arr, low, mid - 1);
   }
   return findFirstZero(arr, mid + 1, high);
}
int main() {
   int n;
   scanf("%d", &n);
   int arr[n];
   for (int i = 0; i < n; i++)
       scanf("%d", &arr[i]);

   int index = findFirstZero(arr, 0, n - 1);
   if (index == -1)
       printf("0");
   else
       printf("%d", n - index);
   return 0;
}
```

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PROGRAM 12: Majority element (Boyer–Moore voting)
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```
#include <stdio.h>
int main() {
   int n;
   scanf("%d", &n);
   int candidate = 0, count = 0;
   for (int i = 0; i < n; i++) {
       int x;

       scanf("%d", &x);
       if (count == 0)
           candidate = x;
       if (x == candidate)
           count++;
```

```
        else
            count--;
    }
    printf("%d", candidate);
    return 0;
}
```

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PROGRAM 13: Find floor in a sorted array (divide and conquer)

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```
#include <stdio.h>
int findFloor(int arr[], int low, int high, int x) {
    if (low > high)
        return -1;
    int mid = low + (high - low) / 2;
    if (arr[mid] == x)
        return arr[mid];
    if (arr[mid] > x)
        return findFloor(arr, low, mid - 1, x);
    int result = findFloor(arr, mid + 1, high, x);
    if (result == -1)
        return arr[mid];
    return result;
}
int main() {
    int n, x;

    scanf("%d", &n);
    int arr[n];
    for (int i = 0; i < n; i++)
        scanf("%d", &arr[i]);
    scanf("%d", &x);
    int result = findFloor(arr, 0, n - 1, x);
    printf("%d", result);
    return 0;
}
```

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PROGRAM 14: Find a pair with given sum in sorted array

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```
#include <stdio.h>
int findPair(int arr[], int left, int right, int x) {
    if (left >= right)
        return 0;
    int sum = arr[left] + arr[right];
    if (sum == x) {
        printf("%d\n", arr[left]);
        printf("%d", arr[right]);
        return 1;
    }
```

```
    if (sum < x)
        return findPair(arr, left + 1, right, x);
    return findPair(arr, left, right - 1, x);
}
int main() {
    int n, x;

    scanf("%d", &n);
    int arr[n];
    for (int i = 0; i < n; i++)
        scanf("%d", &arr[i]);
    scanf("%d", &x);
    if (!findPair(arr, 0, n - 1, x))
        printf("No");
    return 0;
}
```

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PROGRAM 15: Quick sort

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```
#include <stdio.h>
void swap(int *a, int *b) {
    int temp = *a;
    *a = *b;
    *b = temp;
}
int partition(int arr[], int low, int high) {
    int pivot = arr[high];
    int i = low - 1;
    for (int j = low; j < high; j++) {
        if (arr[j] <= pivot) {
            i++;
            swap(&arr[i], &arr[j]);
        }
    }
    swap(&arr[i + 1], &arr[high]);
    return i + 1;
}

void quickSort(int arr[], int low, int high) {
    if (low < high) {
        int p = partition(arr, low, high);
        quickSort(arr, low, p - 1);
        quickSort(arr, p + 1, high);
    }
}
int main() {
    int n;
    scanf("%d", &n);
    int arr[n];
    for (int i = 0; i < n; i++)
        scanf("%d", &arr[i]);
```

```
    quickSort(arr, 0, n - 1);
    for (int i = 0; i < n; i++)
        printf("%d ", arr[i]);
    return 0;
}
```